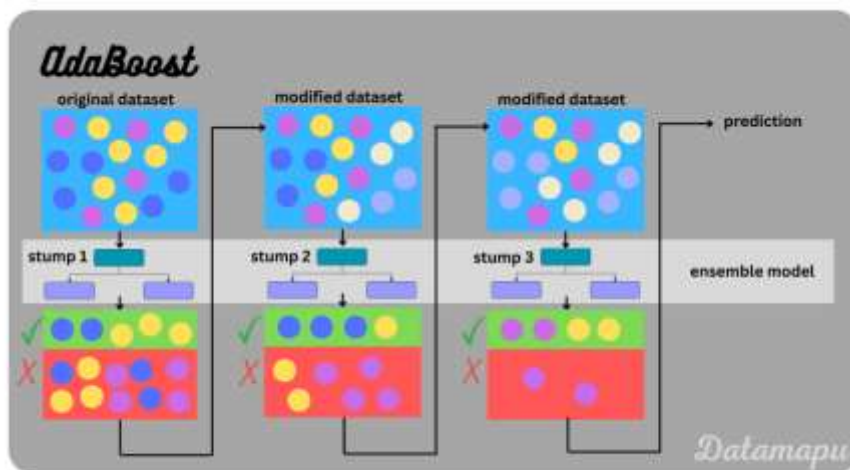


BOOSTING ALGORITHMS

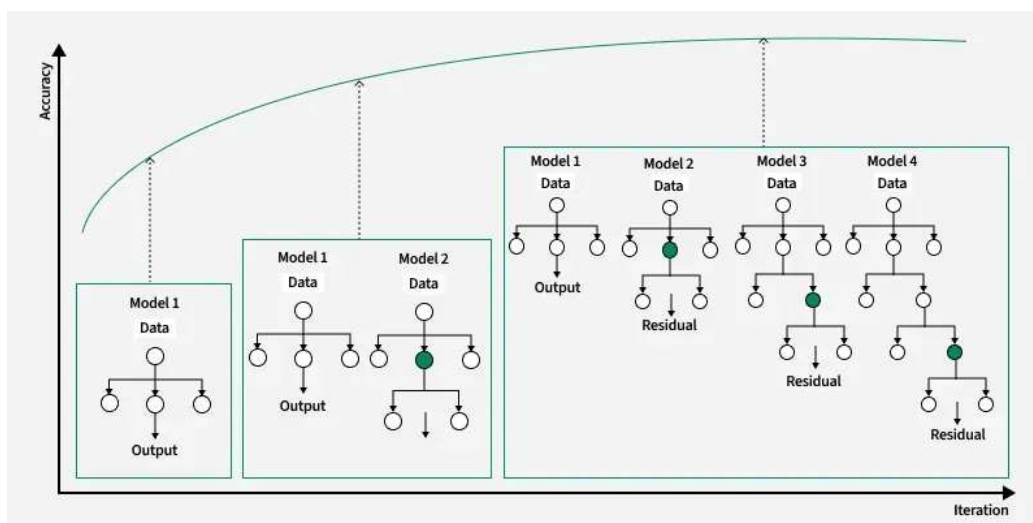
1) ADA BOOST for REGRESSION (Adaptive boosting)

- AdaBoost builds trees sequentially; it assigns weights to training instances and adjusts the weights of data point based on the prediction errors of the previous tree. Harder to predict instances get higher weights, forcing the next tree to focus more on them.



2) LG BOOST (Light Gradient Boosting Machine)

- LG Boost is a gradient boosting framework but is highly optimized for speed and memory efficiency. It uses histogram based algorithms to bucket continuous features into discrete bins which drastically speeds up the split finding process..



3) XG BOOST (Extreme Gradient Boosting)

- XGBoost fits new models to the residuals (the differences between predicted and actual values) of the previous models using gradient descent framework. It includes advanced regularization built directly into the objective function to prevent overfitting.

